

Md Ashraful Islam

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Summary

PhD candidate in CS at UMass Amherst (Advisor: Marco Serafini) specializing in **ML & Data Systems, Agentic AI, and Relational AI**. Published at VLDB'26 (M-TGNN training system with custom CUDA kernels); co-author on Kalypso (relational LLM serving, arXiv'26). Builds high-performance systems with CUDA, PyTorch DDP, and NCCL. Seeking summer internship in ML Systems, Database Systems, or Agentic AI.

Education

- University of Massachusetts Amherst**, PhD in Computer Science Sep 2023 – Current
- **Advisor:** Marco Serafini
- University of Dhaka**, Master's in Computer Science (CGPA: 3.81 – Rank 1st) Jul 2016 – Dec 2017
- **Advisor:** Chowdhury Farhan Ahmed **Thesis:** Weighted Substructure Mining from Transactional Graph Databases (**Graph Mining**)
- University of Dhaka**, Bachelor's in Computer Science (CGPA: 3.89 – Rank 1st) Jan 2012 – Dec 2015
- **Dean's Award 2015** from Faculty of Engineering and Technology, University of Dhaka.
 - **Prof. Dr. Md. Lutfar Rahman Award 2015** from Department of Computer Science and Engineering, University of Dhaka

Research Projects

- PRISM: Scalable M-TGNN Training System** VLDB'26 2023 – Present
- Custom CUDA kernels build a memory computation graph (MCG); diffusion-based propagation eliminates staleness in M-TGNN training at large batch sizes. Published VLDB'26.
 - Extending to multi-node distributed training. (*ML Systems, CUDA, Distributed Training, Graph Learning*)
- Predictive Query over Relational Databases** 2025 – Present
- **Supervised:** Heterogeneous M-TGNN encoder over temporal relational graphs for churn/fraud/conversion prediction without manual feature engineering.
 - **Agentic:** 5-stage pipeline—DB profiling → context graph → task policy → guided row retrieval → SLM head—replacing dedicated feature machines with policy-guided retrieval. (*Agentic AI, Relational Deep Learning, RL, LLM*)
- Kalypso: Relational LLM Serving** arXiv:2607.23815 2026
- Co-designed LLM serving system for semantic query processing pipeline: structured context batching and KV-cache reuse across queries sharing relational context. (*LLM Serving, Database Systems, Inference Optimization*)
- Intrusion Detection via Provenance Graph Learning** 2025 – Present
- Dynamic graph learning over OS-level provenance (processes, files, sockets) to detect stealthy multi-stage intrusions via temporal dependency modeling. (*Security, Graph Neural Networks, Anomaly Detection*)
- Structured Data Mining: Weighted and Utility-Based Pattern Mining** 2017 – 2024
- **Graph Pattern Mining:** Developed efficient weighted frequent subgraph mining algorithms for edge-weighted graph databases with pruning strategies that outperform baseline methods in runtime and scalability (PAKDD'18; *Applied Intelligence* 2024). Extended to single large graphs via WeFreS (ICDM'19) and to utility-based graph mining via UGMINE (ICDMW'21; *Applied Intelligence* 2023).
 - **Sequential Pattern Mining:** Designed weighted frequent sequential pattern mining algorithms with support-weight formulations that capture item significance beyond frequency (*Applied Intelligence* 2022).

Experience

Research Intern, Tursio Inc. — Bellevue, WA Jun – Aug 2026

- Built context-graph enrichment pipeline for an agentic NL-to-SQL system: parsed PDFs, markdown, and schema files into a heterogeneous knowledge graph, reducing hallucinated table/column references.
- Added LLM-in-the-loop result interpretation producing provenance-linked natural language explanations for SQL outputs.

Assistant Professor, University of Dhaka — Dhaka, Bangladesh Sep 2019 – Aug 2023

- Taught Algorithms, Database Systems, Linear Algebra, and ML (30–60 students/class); supervised 6+ MSc/undergrad theses yielding 5 co-authored publications.
- Co-PI on two funded projects: diffusion-based Bangla TTS (University Grant Commission) and Bangla NLP toolkit (Centennial Research Grant, DU).

Lecturer, East West University — Dhaka, Bangladesh Apr 2018 – Aug 2019

- Taught Data Structures, Algorithms, and Discrete Mathematics; developed course materials and mentored capstone projects in ML and data mining.

Skills

Languages: Python, C++, C, Java, SQL, JavaScript

ML/Graph Frameworks: PyTorch, PyG, DGL, CUDA (custom kernels), PyTorch DDP, NCCL

Databases & Systems: PostgreSQL, SQLite, Linux, Git, Docker

Profiling & Optimization: PyTorch Profiler, nvprof, perf

Selected Publications

- Md Ashraf Islam, Hojae Son, Suhaas Kiran Doddagaddavalli Gangadharaiah, and Marco Serafini. Prism: A training system to unlock the potential of temporal graph learning through staleness avoidance. *Proceedings of the VLDB Endowment*, 2026
- Hojae Son, Md Ashraf Islam, Huy Gia Cao, Hui Guan, and Marco Serafini. Kalypso: Relational llm serving. *arXiv preprint arXiv:2607.23815*, 2026
- Mushahid Intesum, Abdullah Ibne Masud, Md Rezaul Karim, and Md Ashraf Islam. Stft-gradtts: A robust, diffusion-based speech synthesis system with istft decoder for bangla. *International Journal of Speech Technology*, 2025
- Md Ashraf Islam, Chowdhury Farhan Ahmed, Md Tanvir Alam, and Carson Kai-Sang Leung. Graph-based substructure pattern mining with edge-weight. *Applied Intelligence*, 54(5):3756–3785, 2024
- Md Tanvir Alam, Amit Roy, Chowdhury Farhan Ahmed, Md Ashraf Islam, and Carson K Leung. Ugmime: utility-based graph mining. *Applied Intelligence*, 53(1):49–68, 2023
- Md Ashraf Islam, Mahfuzur Rahman Rafi, Al-amin Azad, and Jesan Ahammed Ovi. Weighted frequent sequential pattern mining. *Applied Intelligence*, 52(1):254–281, 2022
- Jesan Ahammed Ovi, Md Ashraf Islam, and Md Rezaul Karim. Banep: An end-to-end neural network based model for bangla parts-of-speech tagging. *IEEE Access*, 10:102753–102769, 2022
- Md. Ashraf Islam, Md Towhiduzzaman, Md Tauhidul Islam Bhuiyan, Abdullah Al Maruf, and Jesan Ahammed Ovi. Banel: an encoder-decoder based bangla neural lemmatizer. *SN Applied Sciences*, 4(5):1–15, 2022
- Md Tanvir Alam, Amit Roy, Chowdhury Farhan Ahmed, Md Ashraf Islam, and Carson K Leung. Mining high utility subgraphs. In *2021 International Conference on Data Mining Workshops (ICDMW)*, pages 566–573. IEEE, 2021
- Nahian Ashraf, Riddho Ridwanul Haque, Md Ashraf Islam, Chowdhury Farhan Ahmed, Carson K Leung, Jiaying Jason Mai, and Bryan H Wodi. Wefres: Weighted frequent subgraph mining in a single large graph. In *2019 IEEE International Conference on Data Mining (ICDM)*, pages 201–215. IEEE, 2019
- Md Ashraf Islam, Chowdhury Farhan Ahmed, Carson K Leung, and Calvin SH Hoi. Wfsm-maxpws: an efficient approach for mining weighted frequent subgraphs from edge-weighted graph databases. In *Pacific-asia conference on knowledge discovery and data mining*, pages 664–676. Springer International Publishing Cham, 2018

Full list: scholar.google.com/citations?user=UD5-rcAAAAAJ